

COGNITIVE PSYCHOLOGY RESEARCH GROUP

BioSemi EEG manual

EEG lab manager

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CODE OF CONDUCT

1. Facilities

- EEG Dove kamer (PSI91.67)
 - You can only use these facilities after contacting the lab manager. After approval, the key can then be obtained from the locker near the department kitchen. Ask Agna Marien (agna.marien@kuleuven.be; office 00.37) for the access code (this code changes regularly). Always return the key at the end of your day (even if you are testing multiple days in a row)!
- Two PCs
 - The bitlock code for both PCs is the current year (e.g. 2020).
 - You can login using your own KU Leuven credentials.
 - To ensure computers to be accessible to all, make sure to organize your data files in a single folder. Remove all your files after completing the experiment.
 - Experiment computer (BWSL062) for experiment presentation.
 - This PC has two displays connected to it, one on each desk.
 - Behavioral data is stored on this PC.
 - EEG computer (BWYJ062) for data collection.
 - This PC has one display connected to it.
 - This PC has a LAN connection to the internet.
 - EEG data is stored on this PC.
- BioSemi 64 channels +8 additional channels for EOG, ECG, and EMG.
- Find the list of all equipment, supplies, devices, etc. that are present in the room on the door of the material closet.

Material should not be removed from the room. Bring your own towels to use for your testing. We have no washer/dryer available, so you also have to take your towels back home with you for washing/drying.

Contact the lab manager a.s.a.p. when something is missing/broken, or when supplies are running low.

2. Who can use it?

- Contact the lab manager for the rules and regulations, access to the booking calendar and facilities.
- The BioSemi is available to KU Leuven employees who have experience with the BioSemi system (either prior experience or by following one of our workshops/training sessions).
 - Usage is not free (see 5. Costs).
 - To use the system, you need to complete the BioSemi EEG Workshop beforehand (see 3. Support/training).

3. Support/training

- Before requesting access to the system, read the available documentation on our webpage: <http://copsresearchgroup.be/resources/>
- Before you can use the BioSemi, you must have read this user manual.
- Regularly, a BioSemi EEG workshop will be organized. Attending this workshop is a requirement before being granted access to the equipment. Employees with extensive prior experience with the BioSemi EEG system may be exempted from this workshop. Experience with other EEG systems is not sufficient to be exempted. The lab manager will decide on this.
- Please note that the support we provide does not include help with programming the experiment, analyzing data, or cleaning the equipment. It is only aimed at helping you use the EEG lab and the EEG infrastructure in the correct way.

4. Booking

- Before starting a new study, contact the EEG lab manager and provide them with details of your study:
 - When is it planned to take place?
 - How many participants will you test?
 - Who is running the study (including PI, experiment leader, master thesis students, and/or interns)?
Also provide their contact information.
 - Do you have approval from an ethical committee to conduct this EEG study (+ add ethical approval code)?
 - Do you have prior experience with the BioSemi? Have you already completed the training we organize?
- Provide the lab manager with an @gmail address so that you can be added to the google booking calendar.
- Don't book excessively long periods in advance (prohibiting others from using the lab), and consider testing continuously throughout a day, instead of booking test days interspersed with gaps.

5. Costs

These costs serve to compensate the use of the equipment, consumables and repairs. You pay for each participant you test (i.e., each usage of the material), which also includes pilot testing.

- Members of Brain & Cognition: €15,- per participant.
- Others: please contact the lab manager

6. File Storage

- On the desktops of both PCs, you will find a folder "Experiments". Create a subfolder with your name and store all your files here.
- Always copy your data to an external device and remove it from the PC. The PC's hard disks have limited space and will be cleaned regularly.

7. Cleaning

- You can only use the BioSemi after following the BioSemi EEG workshop, during which you will learn how to thoroughly clean the BioSemi subparts. Researchers exempted from this training still need to obtain instructions from the EEG lab manager on how to clean the equipment, so the same procedure is always used for cleaning by all users.
- The electrodes are extremely fragile. It is therefore crucial to treat them with great care and clean them exactly as instructed.
- Your access to the equipment may be revoked if the cleaning is not satisfactory.
- Please bring your own towels and take them back home with you for washing/drying. Do not leave dirty towels in the lab!
- If you want to offer water to the participant bring your own cups and clean them afterwards. Do not leave dirty cups in the lab!

8. DO's and DON'Ts

- **Always keep one battery charging.** The batteries break if you fully discharge them (“deep discharge”). You cannot harm the batteries by charging them too much.
- Make sure during cleaning **that only the caps and the electrodes get wet.** Make sure that the connectors don’t get wet by wrapping them carefully in a towel or plastic bag.
- If you need a cap that is currently wet you can use a blow dryer, but **use cold air only!**
- Inform the lab manager immediately if something seems to be broken or needs restocking (gel, syringes, adhesive disks , etc.). The EEG equipment is expensive, so handle it with great care!
- Don’t change any hardware or permanent software settings (e.g., refresh rate of the monitor) in our set-ups without asking the EEG lab manager.
- Always book all your test slots via the google calendar.
- Back-up and remove your data from the acquisition machines after each session (e.g., at the end of the day). Computers will regularly be cleaned and loss of data is your own responsibility.
- Always return the key to the locker when you’re not in the lab.
- Place the trashcan outside the room. It will be emptied by the cleaning staff.

USER GUIDE

1. First things to do

These are the first preparations when you come into the EEG lab

1. Switch on both PCs:
 - Experiment PC: BWSL062
 - EEG PC: BWYJ062
2. Use BIT code: the current year (e.g. 2020).
3. Switch on AD box (On/Off button) (Fig 1.).
4. Check whether the battery is charged on the box (Fig. 1), when the battery is very low a red light will turn on.

If the battery is not charged, replace it. Open the black clips at the sides. The spare battery can be found in the opposite corner of the room. Make sure that one battery is always charging (Fig. 2).

(At a later stage you can also check the battery status in ActiView (Fig. 10))

Battery check



Figure 1. AD box on top of battery



Figure 2. Battery charging

2. Before the participant arrives

Make sure that all necessary materials are within reach. You can find everything in the EEG closet in the room. It is the easiest to prepare all the materials on the EEG table, away from the computers.

1. Fill two syringes with gel. Do not fill them completely to the rim (Fig 3).
 - Make sure to clean gel remainders off the tip of the tube before and after filling the syringes to prevent old gel from blocking them (Fig 4).
 - If you only use one of the two syringes you can empty the other back into the tube. **Only do this when the syringe has not been used – don't do this soon as the syringe has touched the participant.**
 - If you test multiple participants in a short timeframe you can re-use the syringes. Make sure to disinfect them properly after each participant (see 7. *Cleaning*).
2. Order and untie the electrodes.
 - If you use the external electrodes, make sure they are within reach, as well as the cleaning scrub to disinfect the participants' face before applying the electrode.
3. Make sure the measuring tape and adhesive tape are within reach.
4. Make sure the head cap you need is dry. You can speed-dry it with the hair dryer but **use cold air only** as not to damage the cap.



Figure 3. Fill the syringes with gel



Figure 4. Clean the tip of the gel tube

3. Upon participant arrival

It is often the first time that a participant participates in an EEG experiment, it may seem scary to them so be sure to explain everything slowly and in great detail.

1. Explain procedure to participant.
2. Give the participant enough time to read the information sheet and sign the informed consent.
3. Ask the participant to remove earrings, hair elastics, etc.
4. In case the participant wears glasses it is easiest to remove them during application of cap and electrodes. After cap preparation and before the start of the experiment it can be placed back on.
If possible, ask participants to wear glasses rather than contact lenses to decrease blinking during the experiment.
5. Measure head circumference from nasion toinion to determine the cap size.
The sizes can be found on the label in the back of the cap.
(e.g., 54 cm → take head cap 52-56 cm; Fig. 5).
6. Both the ears and the label on the back should stick out of the cap.
7. Make sure the cap is properly centered (Fig. 6)
Use the measuring tape to ensure that the Cz (Vertex) is in the center of the head: halfway between the nasion andinion and halfway between the ears.

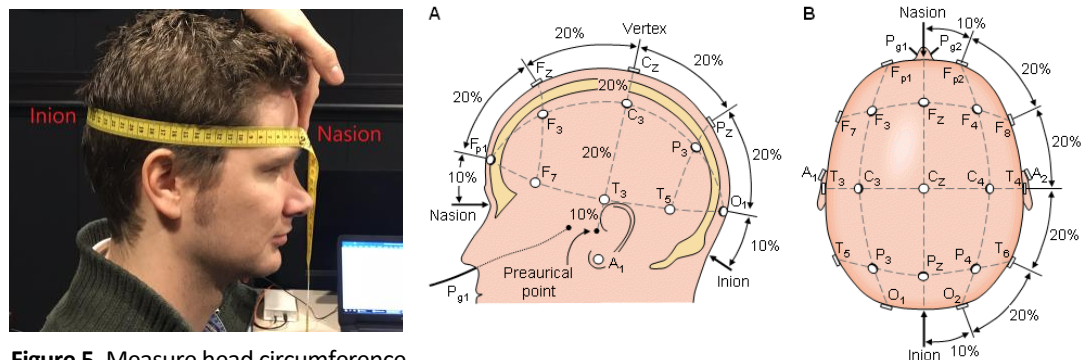


Figure 5. Measure head circumference



Figure 6. Make sure the cap is properly centered

Make sure that the participant is comfortable and in the right position for the experiment, so that they do not have to move much after applying the electrodes.

8. Fill all electrode gaps with superconducting gel (Fig. 7a).
 - Squeeze in a small amount of gel while moving the syringe upwards.
 - Gently fiddle around with the syringe first, as this moves hair aside. Ask the participant for feedback to avoid scratching their scalp.
 - Too much gel can create “bridges” between individual electrodes. Be especially careful around the CMS/DRL electrodes as these are crucial for proper data collection.
It is easier to add extra gel later on than to remove excess gel.
 - You may notice that you easily lose track of which electrode gaps you have already filled. It can help to come up with a system (e.g., start from the back, or if you are with two experimenters you can each start from a side and work your way to the middle).
9. Put the electrodes in the appropriate gaps (Fig. 7b).
 - **Electrodes damage easily, so always handle them with great care.**
 - Electrodes are numbered. Place electrodes in ascending order to avoid tangling the wires. Start at the base of the skull and work your way towards the forehead. Make sure the wires end up in the participant’s neck.
 - Tape the electrodes to the participant’s shoulder to avoid putting stress on the cables.
10. Connect the electrode bundles to the AD box (Brown – A1, Red – B1; Fig. 8).
Handle the connectors with great care, do not force them into the contact.
11. In case the participant wears glasses you can put them back on now.

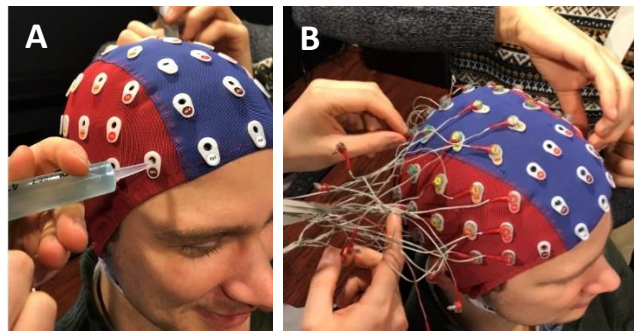


Figure 7. (A) Fill the gaps with gel, (B) Attach the electrodes



Figure 8. Connect the electrode bundles to the AD box

4. Set-up the datafile

1. Start ActiView (Fig. 9).
2. In ActiView, go to the 'about actiview' tab and click *set up configuration file*. In the folder 'configuring' find the 64 channel configuration file.
3. Click *start* and then *Start File*.
4. Browse to your personal folder and choose a name for your data file (e.g. Studyname_SubjXX), enter a filepath in the folder GUI.
5. In the right top corner you can check the battery status of the AD box (Fig. 10). ActiView will give a warning if the battery is lower than 20%.
6. Change decimation according to your needs, e.g. to $\frac{1}{4}$ (gives you a 512 Hz sampling rate; those are fractions of 2048 Hz) and change the trigger format to decimal (so you can see your trigger codes during recording in a useful format).

For more information on how to implement triggers in your script see the example code on page 15.



Figure 9. Set-up your file in ActiView

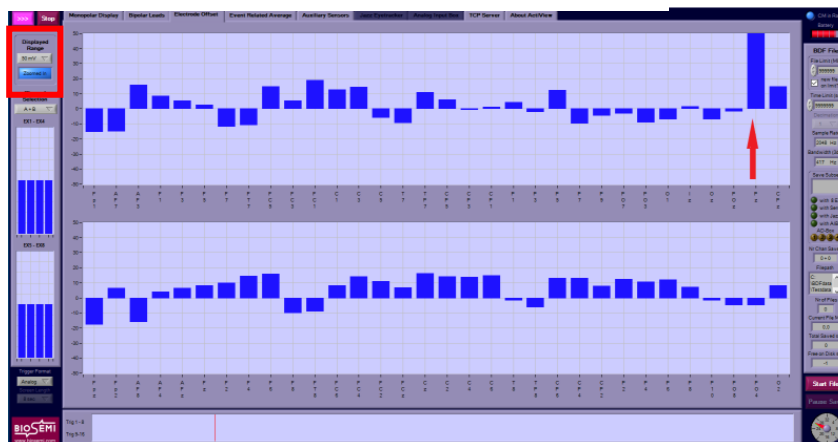


Figure 10. The *electrode offset* tab.

Check the impedance of the electrodes.

7. Switch to tab *Electrode Offset* (Fig. 10).
8. Change the *Displayed range scale* to 50 μV .
9. The impedance of each electrode must be between -20 μV and 20 μV .
Adjust electrodes that are not within these values (e.g., Fz; Fig. 10) by adding/removing some gel, moving hair below the electrode or wiggling the electrode.
 - If the electrode is still far outside the [-20 20 μV] range, the electrode might be broken and needs to be replaced.
In case this electrode is not crucial for your study you can write down the code of the electrode (e.g., Fz) and later contact the EEG lab manager for the back-up electrode bundle.
 - A bad connection of the reference electrode results in a bad signal for all electrodes (Fig. 11a). Check whether the DRL and CMS electrodes are placed correctly and provide a good signal.
10. The cap-label not sticking out of the cap results in a typical signal distortion (Fig. 11b)
11. Show the participant what happens when they blink, cough, swallow, grind their teeth, etc. Instruct the participant to minimize these movements during the experiment.
12. If all electrodes are connected correctly, ActiView should look like figure 12.

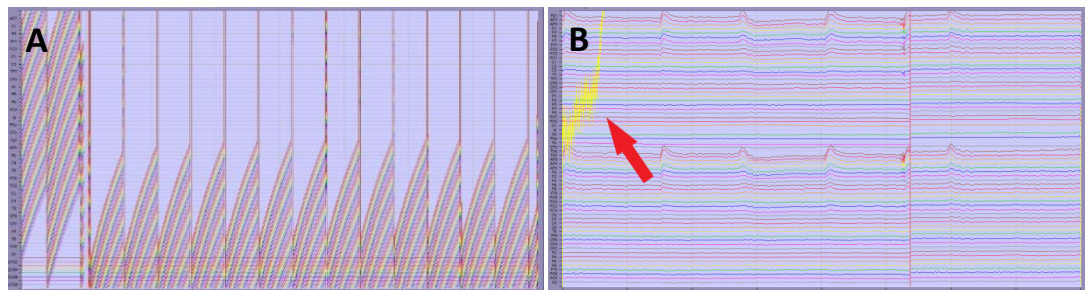


Figure 11. (A) This pattern typically results from a bad connection of the DRL or CMS electrodes. (B) This pattern typically results from the cap label not sticking out of the cap.

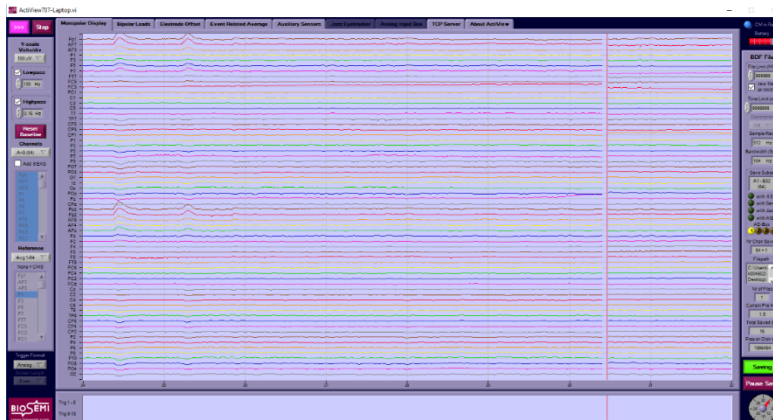


Figure 12. Ideally your data should look something like this, 100Hz sampling rate.

5. Recording

1. **Start recording** (*paused* button; Fig 12 bottom right). When *saving* turns green it means you are recording; Fig. 12). Don't forget this! You wouldn't be the first 😊
2. Before starting the experiment, dim the lights (by keeping the light switch near the door pushed down). Make sure not to dim the lights too much as a high contrast between room and monitor increases blinking.
3. During breaks you might want to turn the lights up (to increase alertness) and ask the participant if they would like to drink some water (to decrease swallowing).
4. Start the experiment.
5. When the experiment is finished, again click *Pause* and *Stop* (Fig 12, top left).
6. The data are saved in the folder selected earlier. Make sure to copy the data from the both computers to a personal drive. Occasionally, the EEG computer will be cleaned.
7. Turn off the AD box.

6. After the experiment

1. Unplug the two electrode bundles from the AD box (Fig. 8).
2. Put the protective caps back on the connectors. Cover the connectors in a waterproof plastic bag. Use an elastic band to close the bag.
The connectors are very sensitive and should not get wet!
3. Remove the electrodes one-by-one from the cap. **Don't pull on the wire but on the electrode itself**, to make sure you don't damage the wires.
Make sure the electrodes never touch metal.
4. Use a towel to drape over the participant's shoulders to prevent gel leaking on the participant's clothes.
5. Once all electrodes are removed, place them in the plastic tub. Remove the head cap and place this in the tub. Place the used syringes in the tub as well.
6. Have the participant sign the attendance list and/or provide their bank details if they participate for course credits or a monetary reward.
7. Show the participant to the shower/sink (see Fig. 13). Bring a towel and shampoo for them if needed.

7. Cleaning

The water supply is quite far from the EEG lab (Fig. 13), so it is easiest to take all equipment that needs to be washed (electrode bundles, caps, syringes, etc.), the cleaning supplies (tub, toothbrush, alcohol solution, etc.) as well as all the belongings of the participant and walk there together.

1. Remove excess gel from the used syringes in the sink.
2. Rinse the syringes, head cap and electrodes (**not the connectors**) in the plastic tub.
3. Use the shower head to thoroughly rinse the head cap with lukewarm water.
4. Rinse the electrodes with lukewarm water in the plastic tub. Carefully brush the gel off of each electrode with a soft toothbrush. Also clean the gaps in the cap with the toothbrush to make sure to remove all gel.
5. Fill the tub again with clean lukewarm water and add a bit of the 70% alcohol solution. Soak the electrodes, the head cap and the syringes for a few seconds to disinfect them. Let soak for a few seconds.
6. Rinse everything again with lukewarm water to rinse off the disinfectant.
7. Return to the EEG room. Let everything dry on a clean, dry towel (Fig. 14) in the EEG room. If you need the cap and electrodes again for the next participant, use the hair dryer too speed up the drying. **Only use cold air!**

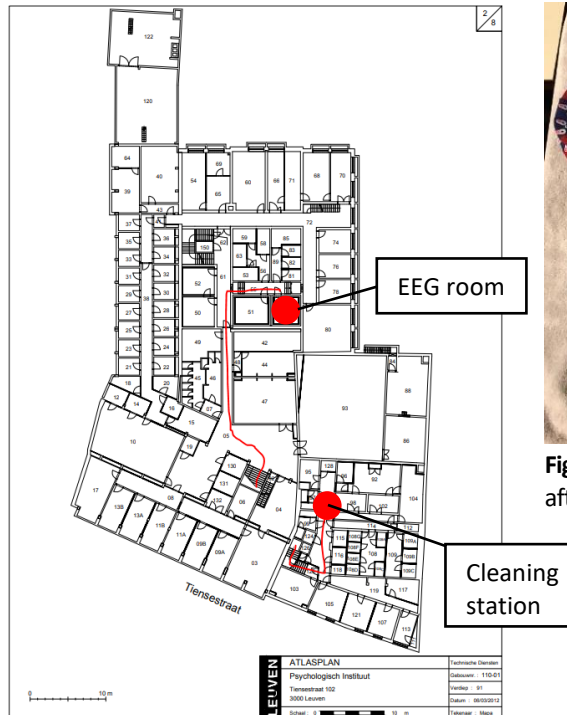


Figure 13. The path between the EEG lab and the cleaning station



Figure 14. Drying of equipment after cleaning.

Import triggers (Python)

```

### Import the serial module ###

import serial

### Initializing code for sending EEG-triggers###

#Here, we define the triggers we want to send throughout the
experiment.
#We use ASCII coding:
# 'A' - 'Z' = 65-90
# 'a' - 'z' = 97-122

# Check what COM via "start" -> "device manager"
serialport = serial.Serial("COM3", baudrate=115200)

# Example list of triggers
Trig_Begin_Experiment    = b'A' #65 --> label in ActiView
Trig_Begin_Block          = b'B' #66
Trig_End_Experiment       = b'C' #67

### Define function to send trigger ###

# We will pass the trigger type (e.g. Trig_Begin_Block) to this
# function whenever we want to send a trigger

def sendTrigger(Trigger):
    serialport.write(Trigger)
    serialport.flush()

### Send trigger to indicate start of a block ###

sendTrigger(Trig_Begin_Block)

### Close serial port at the end of the experiment ###

serialport.close()

```